Lexical Task Heads on Simple Tasks

Schematic project summary

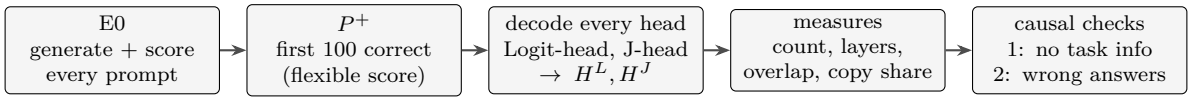
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Marks: ✓ works / supports the hypothesis, ✗ does not, ∼ mixed or task-dependent. Full methods and tables: `proposal-simple.tex`. The complex tasks are summarised in `summary-complex.tex`.

1 Setup in one page

Question	Does the lexical-task-head method of Yang et al. (2026) reproduce on our setup, does Head-JLens (J-head) find heads the per-head logit lens (Logit-head) misses, and which heads should be kept when a task is asked for in several wordings?
Model	Llama-3.1-8B-Instruct (bf16; only model with released Head-JLens matrices).
Readouts	Logit-head: head write $c_{l,h}$ decoded by the unembedding. J-head: head output $z_{l,h}$ mapped through the fitted Jacobian $J_{l,h}$, then decoded.
Simple tasks	14 Todd et al. tasks (antonym, capitals, translation, ...). Rule style: the first 5 of Yang's 6 instruction templates (A–E, incl. the cloze form); example style: 1/5/10/20/30 shots; zero-shot prompt. Raw text, no chat template.
Head criterion	Head is a lexical task head on a prompt if ≥ 1 of its top-10 tokens is in the task's term list; for the style if this holds on $\geq 10\%$ of P^+ (first 100 correct prompts of the most accurate wording).
Term lists	Yang et al.'s (2–22 token ids). Since 11-05 also lists built from the words all heads decode (J-head; E3 below).
Causal check 1	Yang App. G.1: patch the mean head output ($\times\alpha$) at the last token of prompts *without* task info; control = mean of another task at the same heads.
Causal check 2	Yang Sec. 2.4.1/App. H.1.3 in their released script's setting (1-shot): patch prompts that *give* the task but were answered wrongly (baseline 0).
Head sets patched	H^L, H^J, and J-only $H^J \setminus H^L$.



2 Experiments and outcomes

Experiment	Design	Outcome (short)	
E0 — behavioural baseline			
Simple tasks	14 tasks × 11 prompts, every item.	All tasks well above zero-shot; accuracy rises 1→5 shots then flat. Template spread largest on translation, next/prev-item, park–country, synonym.	✓
Scoring audit	Every reply saved; strict (first token) vs flexible.	Translation spread mostly quotes/capitals; first-token score marks any number correct.	✓

E1 — simple tasks (14 tasks × rule/example × 2 readouts)

Experiment	Design	Outcome (short)	
Head sets	100 prompts per style; most accurate template / 5-shot.	Logit-head finds heads on every task. J-head finds more on 23/28 pairs; its extra heads sit in layers 10–19 (Logit-head mostly ≥20). J-head contains all of Logit-head only on 4/28. Example heads mostly inside rule heads (70% shared by Yang's ratio; they report 73%). Counts within 1 head of Yang's Figs. 79/88 in their code's setting; below their Figs. 28/20.	✓
Check 1 (zero-shot)	Means per wording, 15 α values, control mean.	Logit-head: gain ≥10 items on capitals, landmark, park, product (+ currency, next-item, plural with examples). Nothing on antonym, synonym, present–past, prev-item, translations. Best around $\alpha = 4$. Control mostly low (not on park–country, next-item).	~
Check 1, J-head	same	J-head ≥ Logit-head on most tasks; J-only heads have effects of their own.	✓
Cross-style means	Rule heads × example means and vice versa.	Same tasks gain, often as much or more.	✓
Check 2 (1-shot wrong)	Yang's released script, no control.	Gains on all 14 tasks; reproduces Yang's Figs. 57/62 task order. J-head far stronger on next/prev-item with examples.	✓
E1b — heads from several templates (rule style, simple tasks)			
$k = 5$ (all templates)	Heads passing under all 5 templates; both checks.	Better on fact tasks (capitals, park, product) only; worse elsewhere. Beats E1 on ≤5/14 tasks (≤3 by >3 items).	✗
$k = 1$ (union)	Exploratory.	Helps J-head on present–past (0→16 items), synonym, prev-item, control flat. Hurts Logit-head on fact tasks.	~
$k = 3$	Exploratory.	Mostly the E1 heads; behaves like E1.	–
$k = 1$, shared	J-head union, kept only heads in the union sets of ≥ 2 tasks (keeps 15–62 of 19–74 heads per task); both checks.	**Not successful.** Removes most of the union's gains: present–past 16→1 items (check 1), 0.64→0.45 (check 2); synonym 0.21→0.13. Only country–capital rises (0.60→0.69). Shared heads do not carry the task better.	✗
$k = 1$, 4 present–past heads	The 4 heads only present–past's union set contains (L19H10, L19H24, L20H28, L21H9; none in E1); both checks, present–past only.	Check 1: no effect alone (0.01); needs the other 15 heads too. Check 2: 0.63, as the whole 19-head union (0.64); rises with α up to 10, real past forms (60/61 distinct replies); $\alpha = 0$ gives 0.02.	~
Replies, present–past	Each patched reply sorted by kind, per head set and α (J-head, rule means; notebook below).	Shared heads make the model *name* the task ("past tense of assess": 65% of replies in check 1, 73% in check 2 at $\alpha = 10$) without doing it. The 4 unshared heads make it give *a form of the input*: from a bare prompt it repeats the verb (44%), when the prompt shows the task they fix the inflection. Together (union) they give the answer. One task; heuristic reply kinds.	–
Replies, all tasks	Same, every task.	Many patched replies contain the expected answer after a prefix the flexible score rejects ("Capital – Tbilisi", "plural needles", "Country – Russia, . . . "): 19–33% of E1 J-head replies at $\alpha = 4$ on capitals, landmark, antonym, synonym, singular–plural. Not all are answers ("opposite of solid" for *hollow*).	–
Retrieval heads	Yang et al.'s retrieval heads (their code): Definition 1 with the prompt's own answer token as target, on E1's identification prompts; overlap with every lexical set, against layer-matched chance.	Mostly distinct from lexical task heads: 2–9% of each set's heads are retrieval heads, at chance under Logit-head (0.6–1.2×), slightly above under J-head (1.1–1.7×, highest for $k = 5$). None on the translation tasks. Of the 4 present–past unshared heads only L21H9 is one (15% of prompts).	–
4 heads split	Present–past, both checks: L21H9 (the one retrieval head of the 4) alone, and L19H10, L19H24, L20H28 alone.	L21H9 carries the effect: check 2 0.60 alone (4 heads 0.63, 3 heads 0.24); in check 1 it alone makes the model repeat the verb (44% at $\alpha = 10$), as the 4 heads do, and the 3 heads do nothing. A retrieval head that brings the input word forward; the inflection comes from the prompt's example (check 2) or the shared heads (check 1).	~

Experiment	Design	Outcome (short)	
$k = 1$ without retrieval heads	$k = 1$ sets (L and J) minus every retrieval head of the task (any template, either readout): 51/389 L and 84/672 J heads removed, none on translations; both checks, all tasks.	Mostly small changes, but task-dependent. Landmark–country collapses (check 1: L 0.72→0.02, J 0.75→0.13; J replies still name the country, not first). Present–past J loses its gains (L21H9 removed). Gains on synonym J (0.21→0.32), product–company J (0.35→0.48, control 0.10) and prev-item J check 2 (0.12→0.45, at α 9–10, no constant answer). Park–country L: control now above task. Unchanged sets reproduce $k = 1$ exactly. Broad removal: retrieval heads of E1's template and the same readout only would remove 12 L and 51 J heads.	∼
Equal-size sets	Top $N \in \{5, 10, 20\}$ heads by match rate (ties: term count) under each readout and rule (E1, $k = 1/3/5$), only heads matching ≥ 1 prompt (37/420 sets not formed); both checks, own-style means, 9 α ($-10, -4, 2$–$6, 8, 10$); threshold sets compared on the same 9.	**Readouts: J-head's lead is not set size.** Of 25 threshold-set J leads ≥ 0.05, 16 hold at every N, none reverse; none of Logit-head's 4 leads holds (country–capital, present–past reverse). **Rules: mostly size.** Most E1-vs-k differences flip with N; consensus ($k = 5$) below E1 in check 2 (J: antonym, singular–plural, next-item, German). No steady gain with N. Present–past J example: threshold (4 heads) 0.00, top 5 0.87 by adding L15H1 (rate 0.07), control 0; only sets with L15H1 move check 1. L15H1 alone (example means): 0.80 / 0.76 in checks 1 / 2 (top 5: 0.87 / 0.80; the 4 without it 0.00 / 0.57), control 0, none with rule means; J-head reads past participles from it. Control high on park–country (title starting with the country), singular–plural J ex $N = 5$, next-item J ex $N = 20$.	∼
Davidson FV	Davidson et al.'s (2025) instruction function vector, 4 tasks (landmark, product, English–French, present–past): heads ranked by the gain in answer probability when set to their mean on prompts whose instruction is replaced (equiprobable tokens, WikiText, other tasks' instructions), top 20 over the 4 tasks; FV = sum of their mean outputs (inst1–4, 100 prompts), added after layer 11 at the last prompt token; our check 1, 15 α ($\alpha = 1$ is theirs), control.	Peak at $\alpha = 1$ everywhere, nothing at $\lvert\alpha\rvert \geq 3$. Landmark 0.85, product 0.64 (controls 0.06 / 0.12; our best sets 0.79 / 0.46 at $\alpha \approx 3$–4); English–French 0.09 (control 0.03; ours $\leq$0.02); present–past 0.02 (repeats the verb, as our sets). FV heads in layers 13–17 mostly, 0–5 shared with each E1 set. English–French: 8 of the 9 hits are the English input copied where the French differs only in accents. **80 heads** (next 60 by the same score; FV 25–30% longer, cosine 0.88–0.91 with the 20-head FV): still best at $\alpha = 1$; landmark 0.96, product 0.61, English–French 0.19 (11 real translations), present–past 0.07 (real past forms); controls $\leq$0.10. **Set size** (5–1,024 heads, same ranking): no single best. Facts: 5–10 heads at α 3–4 best (landmark 0.95, product 0.87; product control 0.22), at $\alpha = 1$ 40–80 heads (0.95–0.96 / 0.61). English–French: 40 heads, $\alpha = 1$, 0.28 (0.20 real translations). Present–past: 320 heads, $\alpha = 1$, 0.18 (real past forms). All 1,024 heads: facts break (0.01). Size chosen on the test items.	∼
Davidson FV, 14 tasks	As above, heads chosen over all 14 tasks (30 heads; next_item: 12 and 28 correct prompts on two templates, so fewer for its mean); our check 1, all 14 tasks.	Best at $\alpha = 1$ nearly everywhere; FV $\geq$ our best E1 set on 11/14 tasks, controls low ($\leq$0.12). Large gains where our sets do little: singular–plural 0.60 (ours $\leq$0.03), antonym 0.26 ($\leq$0.04), park–country 0.89 (0.39–0.48, our controls 0.29–0.32), English–Spanish 0.10 (real translations). Also above: capitals 0.83 (0.70), landmark 0.89, product 0.64, currency 0.13. Nothing on present–past, prev-item, synonym, English–German (0.01–0.04); English–French 0.09 = copied inputs (accents ignored). Next-item 0.10 vs J 0.17 (J control 0.35).	∼

Experiment	Design	Outcome (short)	
FV heads read	The 30 FV heads (14 tasks), top 10 at the last token: their outputs on correct inst1–4 prompts; overlap with CC-pooled; and the difference mean output (correct prompts) − output on each of the 60 corrupted prompts of the FV score.	Logit-head: noise but 4 heads. J-head, ∼20 readable: answer category (L16H25, L17H4: *cities, currencies, countries, companies*), task domain (L14H13, L15H27, L15H21), relation (L13H22: *opposite, replacement, twice, earlier, next, preceding*), example answers (L17H5, L21H2); ∼1/3 punctuation or noise; answer share ≤0.10. CC-pooled holds 3–11 of the 30 per task (layer-matched chance 4–6); 10 in no CC set (L17H5, format, noise); 34/130 memberships by common words only. Difference: same words for category, domain, relation heads; content under constant punctuation (L14H29 *counterpart, equivalent, translated*; L15H22 *individual, today, previous*); late heads show the item's input. Answer share of the difference finds 5/30 FV heads.	–
Lens FV score	Davidson's head score estimated without patching: per head, the change (task mean − output on each of the 60 corrupted prompts) through J-head (or Logit-head) and the final norm linearised at the prompt, projected on the prompt's own answer token (logit, log-prob, prob); the measured score recomputed on the same prompts.	Checks: means = saved FV, measured = saved score (≤0.001), norm step = finite difference. Over 14 tasks, J-head: Spearman 0.31–0.35 with the measured score (Logit-head 0.14–0.24); 7–8 of the 30 FV heads in its top 30 (Logit-head 4–5, chance 0.9). Finds the late FV heads (L31H2, L31H14, L31H22, L18H29, L17H5, L16H25, L21H2), misses the mid-layer word heads (L15H14 rank ∼950, L13H22 ∼200); its own top 30 mostly layers 24–31.	∼
FV generic heads	FV heads whose J-head tokens are alike across the 14 tasks (mean cosine of token counts), against the 231 equally generic heads of layers 13–31: attention at the last token (split by prompt part), mean output per task (J-head-transported), change by the instruction, per-task score.	6–7 generic: L14H29, L15H5, L14H11 (*...\n, :\n, ;\n*), L13H31, L18H22, L31H14 (Logit-head: *, (a in*), L15H7 (*instead, named, Wikipedia*). Near-constant output (cosine 0.80) and small instruction change (0.17) are shared by the equally generic heads (0.78, 0.18); what differs is attention off BOS (0.25 vs 0.88). L14H11, L15H5, L13H31 put 40–62% on the answer slot (newline, *A*, final colon), L31H14 99% on the last token; the task FV heads read task words (0.21–0.32) and the input (0.14–0.29). Generic decoding with attention off BOS: 9/608 heads, 5 FV (score ranks 9–27).	–
FV split	Davidson's 30-head FV in our check 1, split: the 6 generic heads (generic6), the other 24 (task24), and the 30 minus 6 random non-generic heads (3 draws); $\alpha \in \{0, 1, 2, 3, 5, 8\}$; reply length and line-break stops; all30 = saved run (7/168 rows differ by ≤0.03, batching).	generic6 alone: nothing (0.03). **task24 beats the full FV**: 0.58 vs 0.33 (best α per task), ≥ all30 on all 14, best at α 2–3 (0.54 at fixed $\alpha = 3$); translations 0.39–0.70 (FV ≤0.10), antonym 0.61, synonym 0.35. Dropping 6 random task heads: 0.25–0.30, still peak at $\alpha = 1$. The generic part is most of the FV's norm (9–15 of 11–16), the same direction on every task (cosine 0.96), orthogonal to the task part; scaled up it turns replies into *://*, *.swing*, *1. 2. 3.* on every task.	✓

E3 — head sets from decoded words (J-head only, rule style; check 1 only, rule means averaged over the 5 templates, control mean, $\alpha \in \{1, 2, 4, 8\}$, best α reported. Numbers are means over the 14 tasks; E1 J-head on the same α: 0.19.) Words: a decoded top-10 token, stripped and lowercased, kept if alphabetic, ≥ 3 letters, not a stop word, common English (`wordfreq` Zipf ≥ 3); *share* = fraction of (prompt, head) pairs over all 1,024 heads whose top-10 holds it; *task words* = share in the task − mean share in the other 13; *common words* = highest minimum share over the 14 tasks. Heads: Definition 1 with the new list ($k = 10$, $\rho = 0.10$) unless stated. Only P^+ (correct) prompts.

Top decoded words	Top 40 words per task over all heads, no baseline.	Task names and answer types lead (*opposite*, *city*, *currency*, *French*); generic words on every task (*what*, *which*, *maybe*, *https*). Yang's list words only partly among them.	–
Common + task, 4 tasks	20 common and 20 task words from the best template; heads on the best template; both readouts; antonym, present–past, capitals, product.	Task list J best (capitals 0.59, antonym 0.22); common lists ≈ 0 except product under Logit-head (0.32).	∼
All 14 tasks	Same, words counted on all 5 templates.	Task J 0.10, common J 0.07: below E1.	✗
List length	5, 10, 15, 20 words.	Shorter is better: task J 0.22 (5 words) → 0.10 (20). Common ≤ 0.07.	∼

Experiment	Design	Outcome (short)	
Heads on pooled prompts	Definition 1 on the ≈500 prompts of the 5 templates pooled instead of the best template; also Yang's list.	Helps every list: task J 5 words 0.25; common J 0.11–0.17 (synonym 0.58 with 10 common words, control high); Yang's list pooled 0.21.	∼
Combined 5+5 (2×2)	5 task + 5 common words; words and heads each from the best template or pooled.	**Both pooled = CC-pooled**: 0.45, 95–137 heads, above E1 on all 14 tasks; carries tasks E1 does not (antonym 0.54, singular–plural 0.64, next-item 0.43, translations 0.25–0.32). Present–past 0.03. Controls: synonym 0.26, park 0.17, singular–plural 0.12, next-item 0.12. Words pooled, heads best 0.30; words best, heads pooled 0.31; both best 0.18.	✓
Random sets	Best set per task so far vs 5 layer-matched random sets of its size.	Best 0.49, random 0.03–0.07. Single draws high on antonym (0.45) and landmark (0.42); product 0.16–0.39 on all draws.	✓
Strict consensus	5+5 and 10+10 lists, head must pass on all 5 templates, top-5 tokens, $\rho = 0.15$ (run at $\rho = 0.05$ stopped); also its top 20 by rate.	0.15–0.20. Best only on country–capital (0.78).	✗
Neutral cut (CC-pooled-clean)	J-head readings on 1,000 WikiText-103 passages (lengths matched); the 50 words heads read most there removed before choosing task and common words; 5+5 and 10+10, pooled.	5+5: 0.38, 10+10: 0.33. Above CC-pooled on 8/14 (facts, translations, morphology: landmark 0.97, French 0.40, German 0.49, singular–plural 0.72, present–past 0.14). Loses antonym, synonym, next/prev-item: they need the generic words. New common words: *named, therefore, question, answer, very.*	∼
Task minus WikiText	Top 10 task words after the cut; no cross-task contrast, no common words.	0.21. Landmark 0.00 (patched replies start "Country"), translations name the language.	✗
Group contrast	5 groups (word relations, morphology, facts, sequences, translations); task words = share − max share of the other tasks of its group; + 5 top WikiText words (*which, mph, http, his, what*).	0.32. Best antonym of all (0.70). Landmark and park collapse (0.01 / 0.08): *country* is shared within the group and removed.	∼
Match-rate ranking	CC-pooled heads ranked by pooled match rate; top 1, 2, 5, 10, 20, 40; layer-matched random at 10, 20.	Small sets work only on capitals (L16H25 alone 0.88) and landmark (40 heads 0.93). Means: 5 heads 0.19, 20 heads 0.21, 40 heads 0.25.	∼
Relation heads	Head's top-10 holds both a term-list token and a token of the prompt's own query.	Rare: $\rho \geq 0.10$ for ≤ 7 heads per task, none on 10 tasks. Not patched. L16H25 decodes *city*, not the relation.	–
Attend + decode	L16H25 pattern: at the last token it attends to *capital*, *country* in the instruction and decodes *city*. Score = (share of last-token attention on task words in the instruction, outside the query) × (share of prompts with a task word in the J-head top 10); task words = Yang's list ∪ 10 top task words; eager attention, pooled prompts. Top 1, 2, 3, 5, 10, 20; layer-matched random at 5, 10.	Few heads carry fact tasks and plurals: capitals 0.88 (L16H25 alone), landmark 0.79 (5 heads), product–company 0.72 (20 heads, control 0.09; best of all), singular–plural 0.68 (20). Antonym 0.25 (5). Nothing on translations, next/prev-item, present–past. Park: control nearly as high. L16H25 ranks first on all 5 fact tasks; pattern in layers 13–17. Means: 5 heads 0.21, 20 heads 0.25 (match rate 0.19 / 0.21); random 0.04.	∼

E4 — held-out test of the E3 sets (J-head, rule style, E3 designs frozen. Items of each task split in two halves; words, heads, rankings and means chosen on one, patched on the other, both ways. Both checks, control mean in both; 10 α (the 9 of the equal-size sets + $\alpha = 1$); α cross-fitted: the best on the other fold. Accuracies only, no intervals. Numbers: means over the 14 tasks, check 1 unless stated.)

CC vs E1	CC-pooled (97–154 heads) and the E1 set (7–50), both rechosen on each half.	CC 0.44 vs E1 0.20 (E3: 0.45 vs 0.19). Above E1 on 13/14 (not product–company); still carries antonym, singular–plural, translations; next-item lead small; present–past 0.02.	✓
Random	5 layer-matched draws per size, at the CC set's α.	0.02 at every size (product–company 0.15); check 2: 0.11–0.13.	✓

Experiment	Design	Outcome (short)	
Decoded vs causal	Same number of heads ranked by Davidson's FV score on the choosing half: FV-20, FV-40, FV-N_t.	**FV better at every size:** N_t 0.59 vs CC 0.44; 20: 0.38 vs 0.22; 40: 0.43 vs 0.26. CC ahead only on prev-item and translations at N_t (German 0.39 vs 0.01). FV carries present–past (0.85). FV best at α 1–2, CC 2–3. Davidson's own 30-head FV ($\alpha = 1$): 0.39.	✗
Check 2	1-shot prompts answered wrongly.	Sets close: E1 0.31, CC 0.33, FV-N_t 0.39; controls 0.12–0.13, as large as the task mean on next/prev-item; 11–34 items per fold on 5 tasks.	∼
Secondary	Fold 1 only, best α: CC-clean, attend + decode 20/40, FV-univ 20/40 (FV heads ranked over all 14 tasks, Davidson's choice).	CC-clean 0.39, A+D 0.28 (product–company 0.80), FV-univ-20 0.63, highest of all sets.	–
E4b — more wordings, held-out wordings (design in `proposal-simple`, Sec. 16.3)			
New wordings	30 per task: 3 forms (cloze, Q–A, label) × task named / not named; GPT-5.5, script checks, hand check; then heads from $m \in \{1, 5, 11, 23\}$ wordings at a fixed prompt count, tested on held-out wordings.	84 cells × 5 generated (+ 2–5 spares each). **Pending:** hand check; nothing run yet.	–
E6 — heads from what the instruction adds (J-head, rule style; E4's split, CC rule, patching and cross-fit, frozen)			
Difference readings	Per E4 identification prompt (inst1–4; inst0 has no instruction), clean z minus the mean of two corrupted versions, instruction replaced by Davidson et al.'s length- and log-prob-matched text (equiprobable tokens, WikiText; other-task instructions left out: related tasks share processes), top 10 under J-head. Sets: dcc (CC rule), dccT (5 task words only), top 20/40, answer heads removed (A). E4's CC-20 re-patched = E4 exactly.	The common words do not cancel (*what, https, http, mph*): dcc ≈ CC size (84–128 heads), dccT 21–68. Check 1: dcc 0.18, dccT 0.16, dcc20/dccT20 0.22, dcc40 0.22, dccT40 0.20; CC 0.44, CC-20 0.22, CC-40 0.26, FV-N_t 0.59, random ≤0.04. Ahead only on capitals (0.80–0.84 vs CC-20 0.77) and product (0.60–0.62 vs 0.51); lose antonym, synonym, translations, sequences, singular–plural. dcc keeps ∼half of CC's heads; the dropped ones decode the words with and without the instruction. Check 2: all 0.32–0.34 (CC 0.33). Answer removal: little.	✗
E7 — most active words, then an attention filter (J-head, rule style; E4's split, word rules, CC head rule, patching and cross-fit, frozen. Motivation: word-decoding heads outside the FV set attend to BOS (∼0.80), FV task heads to the prompt (0.25–0.39).)			
Lists and sets	Raw list: the task's 10 most active words (share over the 500 pooled prompts of the choosing half), no cross-task contrast. Clean list: the same after removing the 50 top WikiText words. Heads: CC rule (≥10% of the 500). Filter: drop heads whose mean last-token attention on BOS over the task's prompts is > 0.5.	Raw lists ≈ half task words, half generic (*what, which* on all 14, *http* on 7); clean lists task words only. Folds agree on 9–10/10 words (next-item 4/10). Sets: raw 72–135 heads, clean 58–101; the filter keeps about a third or less (raw 17–43, clean 20–39; median BOS of the unfiltered sets 0.57–0.77).	–
Check 1	Raw, raw + filter, clean, clean + filter; random-drop (as many heads as the filtered set, drawn from the unfiltered set) and layer-matched random, 5 draws each, at the filtered set's α.	Raw 0.37 → **raw + filter 0.40** with 32 heads, between FV-20 (0.38) and FV-40 (0.43) and above them on 8–9/14 (CC 0.44 with ∼120, CC-20 0.22, CC-40 0.26, FV-N_t 0.59); clean 0.18 → 0.20. **Random-drop 0.09–0.10**, layer-matched 0.01–0.02: the filter keeps the heads that carry the task. Filter better on 10/14 for both lists; loses antonym (0.62 → 0.46) and Spanish (0.44 → 0.12). Raw + filter above CC on 9/14 (singular–plural 0.88, capitals 0.87, German 0.54, product 0.46, present–past 0.21 vs 0.02) and above FV-N_t on prev-item and translations; landmark collapses on every E7 set (0.00–0.05 vs CC 0.92; replies *Country: Germany*, *A Journey Through Time*). Controls ≤0.04.	✓

Experiment	Design	Outcome (short)	
Check 2	The four main sets, control mean.	Raw 0.31, raw + filter 0.34, clean 0.29, clean + filter 0.32 (CC 0.33, FV-N_t 0.39). Prev-item raw + filter: control 0.49 above task 0.15.	∼
E8 — every combination of five head filters (J-head, rule style; E4's split and protocol, $\alpha \in \{1, 2\}$ only, cross-fitted; references re-cross-fitted over α 1–2: CC 0.40, FV-20 0.33, E7 raw + filter 0.31, FV-N_t 0.59. Report: `paper/e8-filter-combos`.)			
Filters and sets	D: J-head top 10 holds a word of the task's 30 most active decoded words on ≥10% of prompts (E7 word rules; tokens with leading-space capitalised/upper forms added); B: BOS attention ≤0.5; T: attention on the instruction's task words (Yang terms + E4 specific words) ≥0.05; I: on the query ≥0.05; R: on the rest ≤0.5. All 31 AND-combinations, no size cap, no controls.	First run with E7's 10-word list and Yang's expansion missed the " Capital" tokens the lens decodes (L21H2: " England" on 488/500 prompts, D rate 0). Fix + 30 words (chosen on FV-20 overlap of the choosing halves, no patching): all 16 D sets rerun; sets about 2× larger (DB 32 → 65 heads).	–
Check 1	All 31 sets; fold-1 pick read on fold 2.	**T 0.55** (154 heads; fold-1 pick, fold 2 0.55), **DB 0.51** (65; first run 0.31), DT 0.49, DBT 0.49 (38), D 0.41. DB above CC and FV-20 on 9/14, below FV-N_t on 10/14; T wins the facts, DB word relations, present–past (T has 13 heads there) and French. B raises D (+0.10) but lowers T (−0.09); adding I or R lowers all 15 sets. Next/prev-item ≤0.11 on every set. DB holds 8.5 of FV-20 (chance 1.3).	✓
Check 2	Same sets.	Close: T 0.40 ≈ FV-N_t 0.39, DB 0.32, CC 0.34; T sets' next-item gains matched by their control.	∼
E10 — one shared 30-head set for all 14 tasks (J-head, rule style; E8's scores, split and protocol, $\alpha \in \{1, 2\}$ cross-fitted; each task still patched with its own means. Aim: a Davidson-style shared set, for a direct comparison. Report: `results/e10_shared/report.md`.)			
Sets	Each of E8's five per-head scores combined over the 14 tasks: average (`mean`), weakest task (`worst`: lowest D/T/I, highest B/R), average within-task rank (`rank`). Per combination, the heads' ranks on its filters averaged, best 30 kept; no thresholds (on the weakest task they leave <30 heads in 27/31 combinations). 93 sets per fold.	Folds share ∼29/30 heads (`mean`), ∼25 (`worst`). Overlap with Davidson et al.'s 30 FV heads up to 13 (DBT, DT; chance 3–4), 0 for sets with R; mostly their layer 13–16 heads, rarely the late ones.	–
Check 1	All 93 sets; fold-1 pick read on fold 2.	**rank_DBT 0.47** (fold-1 pick, fold 2 0.47; E8 per-task DBT 0.48), mean_BT 0.44, rank_DB 0.43, mean_DBT 0.42: about equal to E8's per-task sets of the same combination (DBT 0.49, BT 0.45, DB 0.51), above Davidson's shared FV (0.33, residual-stream FV, other items), below FV-N_t 0.59. T alone loses most (0.35 vs 0.55). `worst` lowest (best 0.36); R or I combinations ≈0.01–0.2. Shared sets gain on facts and present–past (rank_DBT 0.16 vs 0.00) and lose antonym (0.08 vs 0.61), synonym and translations. Controls ≤0.05; no constant replies.	✓
Check 2	Same sets.	Best sets 0.36–0.38, T 0.41: equal to E8 per task (BT 0.38) and FV-N_t (0.39).	∼
E11 — completing the comparison (J-head, rule style; E8 protocol, $\alpha \in \{1, 2\}$ cross-fitted, both folds and checks; sets from existing rankings and random-draw rules only. Synthesis: `paper/head-sets-simple`.)			

Experiment	Design	Outcome (short)	
Sets	FV-univ-N and E10 shared DBT-N at 5–40 heads; FV-univ-20 minus its generic heads; random shared sets (layers of FV-univ-20 / E10 DBT); per task random layer-matched to DB, DBT, T and drawn from D at DB/DBT size; E4's random sets in check 2; Davidson FV-30 (their intervention) and FV-30 minus its 5 generic heads (25), both checks.	Check 1: **FV-30 without generic heads 0.63** at $\alpha = 2$ (one set for all tasks; FV-N_t 0.59, T 0.55), carries every task type but sequences; FV-30 at $\alpha = 1$ 0.39, collapses at $\alpha = 2$. FV-univ: 5 0.21, 10 0.23, 20 0.40 (fold 1 0.48, fold 2 0.33), 30 0.52, 40 0.53; FV-univ-20 minus generic 0.51. E10 DBT: 0.15 / 0.22 / 0.31 / 0.47 / 0.52. Random shared 0.01; random from D 0.13 (DB size) and 0.06 (DBT size); layer-matched 0.01 (T: 0.18, overlapping draws). Check 2: all 0.25–0.41, random 0.06–0.18. Caveat: generic heads removed after the FV split on the same items.	✓

E12 — generic heads found on each choosing half (J-head; E11 protocol, both folds and checks, $\alpha \in \{1, 2\}$ cross-fitted. Report: `scripts/e12_report.py`; `paper/head-sets-simple`, App. C.)

Sets	J-head generic score of the fold's FV-30 heads on the choosing half only; generic = heads above the largest drop in that score (rule fixed before looking): 9 heads in fold 1, 8 in fold 2 (the five old ones plus L15H7, L16H29, L12H9, and L28H10 in fold 1). FV-30 minus them (21/22 heads, their intervention); FV-univ-20 minus them (15/15, head by head).	Check 1: **FV-30 without generic heads 0.63 → 0.49**; FV-univ-20 minus generic 0.51 → 0.42. Check 2: 0.41 → 0.38, 0.39 → 0.38. The loss is on translations, present–past and prev-item; facts unchanged. E11's best set was not cleanly held out.	✗

E13 — every rule at 20, 40, 60 and 80 heads (J-head, rule style; E4 split, check 1 only, $\alpha \in \{1, 2\}$ cross-fitted, control mean; one set per task and one shared set. Synthesis: `paper/head-sets-simple`, Secs. 4–5; results `results/e13_capped`.)

Lists and sets	One word list per task and fold, for D and T: J-head top 10 at the last token of the 5 rule templates' correct prompts (choosing half); a word = alphanumeric (any script), not one of 18 stop words, ≥3 characters if Latin; the 50 top WikiText words removed; first 30 by share. T = attention to list words in the instruction (Yang terms no longer added). D, B, T and their combinations ranked by averaged ranks (shared: E10's rank rule); Causal = Davidson's score per task / averaged; DBT and Causal − generic (two-group split of the FV-30's generic scores on 5 templates: E12's heads); Davidson's summed vector ± generic (shared); Yang terms (heads with rate >0 only: 45–80 heads on 7 tasks at 60/80); random layer-matched to DBT (5 draws). 128 head-patched sets per task and fold.	Per task (20/40/60/80): **DBT 0.43/0.55/0.57/0.55**, BT 0.42/0.54/0.58/0.55, T 0.31/0.45/0.48/0.54, DB 0.44/0.45/0.36/0.32 (larger sets describe the query), Causal 0.33/0.41/0.46/0.53, Causal − gen. 0.38/0.46/0.50/0.51, Yang 0.18–0.21. Shared: **Causal − gen. 0.50/0.55/0.64/0.69** (best at every size), Causal 0.40/0.53/0.60/0.65, DBT 0.33/0.46/0.61/0.55, summed FV 0.24/0.41/0.41/0.42, summed FV − gen. 0.49/0.49/0.49/0.52. Removing generic heads helps Causal and the summed FV, lowers DBT. D adds nothing to BT. Translations: per-task filters, shared Causal from 40 heads, never per-task Causal; sequences ≤0.36 everywhere. Random ≤0.01, controls ≤0.05.	✓

E20 — where E13's heads look, by Bai et al.'s token types (diagnostic, no patching; choosing half of each fold, the 5 rule templates' correct prompts, last-token attention. Idea: Bai et al. (arXiv 2401.11323v4) find that demonstrations' information collects at template tokens, punctuation and stopwords. Results `results/e20_bai_attention/report.md`.)

Experiment	Design	Outcome (short)	
Categories	E13's "rest" split, automatically: last token (self); the Q–A scaffold " Q:" and "\nA" (template; none in cloze templates); punctuation; NLTK stopwords (function); other instruction words (content). Task words, query and BOS as E13 (reproduced exactly; T sets 56/56). Compared: mean masses of E13's 40/80-head sets, generic heads, all heads; Spearman over 1,024 heads with Davidson's score; overlap with Causal-N of a top N on each category.	Causal heads do *not* look more at template tokens or punctuation than DBT/BT: filter sets attend more to every prompt category, causal sets more to BOS (0.44 vs 0.33) and to the last token (0.12 vs 0.08). Generic heads: self 0.26, template 0.22 (Q–A: 0.28), the answer slot. All categories correlate weakly with the causal score (per task $\rho \leq 0.09$; shared $\leq$0.23), template and punctuation no better than task words. Ranking on the whole instruction (no word list) shares 26/40 heads with T and as many with Causal-40 (11.3 vs 10.6; shared 14.5 vs 17); on present–past more (12 vs 4–8).	✗
E21 — E20's token types as head rankings (J-head, rule style; E13's protocol exactly: E4 split, check 1, $\alpha \in \{1, 2\}$ cross-fitted, control mean; 40 heads, per task and shared. Results `results/e21_bai_rules/report.md`.)			
Sets	New scores from E20's attention split (choosing half, 5 templates, last token, higher first): L last token, M template markers " Q:"/"\nA" (0 on cloze), S punctuation + NLTK stopwords of the instruction (Bai et al.'s stopword type), N whole instruction (T without the word list). Each alone, with B and with DB, ranks combined as E13 (shared: E10's rule). Builder reproduces E13's T/BT/DBT-40 exactly. No random sets (E13's stay the reference).	Per task: **DBS 0.60** (folds 0.61/0.60; E13 DBT-40 0.55, 9 of 14 tasks up, 5 down), the best per-task set at 40 heads and above every E13 per-task set up to 120 heads except Causal-120 (0.61); N 0.53 vs T 0.45 (no word list; present–past 0.45 vs 0.00); BN 0.55, DBN 0.54, BS 0.53, S 0.39; M 0.10, L 0.01 (= random). Shared: none beats Causal − gen. 0.55; DBM 0.53, DBL 0.51 (morphology 0.86–0.88), DBN = DBT 0.46, N 0.31. Controls $\leq$0.05 (S, N alone highest).	✓
E22 — every combination of the seven filters (J-head, rule style; E13's protocol exactly: E4 split, check 1, $\alpha \in \{1, 2\}$ cross-fitted, control mean; 40 heads, per task and shared. Results `results/e22_all_combos/report.md`.)			
Sets	All combinations of D, B, T (E13) and L, M, S, N (E21) except those holding N with T or S (N already counts their positions): 79, of which 19 were run in E13/E21 (rebuilt exactly) and 60 are new. Ranks combined as E13/E21.	Per task: **DBMN 0.63** (folds 0.63/0.63), DBTM 0.61, DBTMS 0.61, DBS 0.60, DBT 0.55; DBMN is above every earlier per-task set at any size (E13 Causal-120 0.61). Shared: **DBTLS 0.60** (folds 0.60/0.61), DBTLM and DBTM 0.59, DBMN 0.58: above Causal − gen. at 40 (0.55), below it at 80–100 (0.69–0.71). Mean gain of adding a letter (with vs without, every pair; per task/shared): D +0.24/+0.32 (nearly every pair), N +0.13/+0.05, T +0.10/+0.08, B +0.06/+0.09, S +0.06/−0.02, M −0.05/−0.03, L −0.09/−0.05. M helps only next to DB with T or N (DBN→DBMN +0.09, DBT→DBTM +0.06) and hurts next to S (DBS→DBMS −0.09). Without D, every combination holding L or M is $\leq$0.35 per task. Sequences stay low in the ten best ($\leq$0.19 per task). Controls $\leq$0.05. The best are the best of 79 on the test items, so their accuracy is optimistic by selection.	✓
E23 — the 15 best E22 combinations per scope at every size (J-head, rule style; E13's protocol exactly; 20, 60, 80, 100, 120 heads, 40 from E13/E21/E22; E13's own filter combinations not rerun. Results `results/e23_best_sizes/report.md`; Table 1.)			

Experiment	Design	Outcome (short)	
Sets	The 15 best of E22's 79 at 40 heads, per task and shared chosen separately, built by E22's rule at every E13 size (the rebuilt 40-head sets equal the patched ones); 145 new sets per task and fold. Chosen on the same test items, so their curves are optimistic.	Per task: **DBTMS 0.68** at 100–120 heads, DBTLS 0.67 at 80, DBMN 0.66 at 60; 11 of the 15 peak above the earlier per-task best (Causal-120 0.61; T-120 0.58, BT-60 0.58). Combinations with S keep rising to 100–120 heads; DBT falls after 60 (0.41 at 120); DBMN and DBTM peak at 60. At 20 heads DBS and DBTS lead (0.47). By task type the filter sets win translations (0.53–0.69 vs Causal 0.26) and most word relations; Causal keeps sequences (0.43 vs ≤0.25) and morphology (0.89 vs ≤0.84). Shared: Causal − gen. still reaches the top (0.71 at 100); DBTMS ties it at 120 (0.71), DBMN 0.70 at 80; the new sets lead only at 40 (0.60 vs 0.55) and tie at 20 and 60. Shared curves are noisier (fold gap up to 0.14, DBM). Causal − gen. keeps sequences (0.41 vs ≤0.27). Controls ≤0.05.	✓

Table 1: E23: check-1 accuracy (held-out items, α cross-fitted, mean of folds and of 14 tasks) of the 15 best E22 combinations per scope at every size, ranked by their 40-head accuracy, and E13's reference sets. Rows 1–15 were chosen on these items at 40 heads. Bold: best of the column at two decimals. D decodes task words; B low BOS; T task words; L last token; M template markers; S punctuation + stopwords; N whole instruction. Random: layer-matched to DBT.

per-task sets (one set per task)

Set	20	40	60	80	100	120
1. DBMN	0.36	**0.63**	**0.66**	0.65	0.65	0.62
2. DBTM	0.41	0.61	**0.66**	0.64	0.60	0.57
3. DBTMS	0.36	0.61	0.65	**0.67**	**0.68**	**0.68**
4. DBS	**0.47**	0.60	0.62	0.63	0.62	0.58
5. DBTS	**0.47**	0.57	0.62	0.61	0.64	0.64
6. DBTLS	0.43	0.56	**0.66**	**0.67**	0.66	0.65
7. BTS	0.43	0.56	0.60	0.62	0.62	0.64
8. DS	0.41	0.56	0.61	0.62	0.63	0.64
9. DTS	0.43	0.55	0.59	0.61	0.62	0.64
10. DBT	0.43	0.55	0.57	0.55	0.48	0.41
11. DBLN	0.46	0.55	0.62	0.59	0.59	0.55
12. DBLS	0.41	0.55	0.62	0.63	0.65	0.63
13. BN	0.42	0.55	0.57	0.59	0.58	0.59
14. DBN	0.43	0.54	0.58	0.58	0.55	0.53
15. DLN	0.41	0.54	0.58	0.57	0.58	0.58
Causal	0.33	0.41	0.46	0.53	0.56	0.61
Causal − generic	0.38	0.46	0.50	0.51	0.53	0.58
T (E13)	0.31	0.45	0.48	0.54	0.57	0.58
BT (E13)	0.42	0.54	0.58	0.55	0.57	0.57
random	0.01	0.01	0.01	0.01	0.01	0.01

shared sets (one set for all 14 tasks)

Set	20	40	60	80	100	120
1. DBTLS	0.41	**0.60**	0.62	0.67	0.64	0.65
2. DBTLM	**0.50**	0.59	0.64	0.60	0.58	0.62
3. DBTM	0.37	0.58	0.62	0.66	0.69	0.68
4. DBTLMS	0.37	0.58	**0.65**	0.65	0.67	0.68
5. DBMN	0.35	0.58	0.64	**0.70**	0.68	0.68
6. DBMS	0.34	0.58	0.62	0.60	0.67	0.69
7. DBLMN	0.43	0.57	0.58	0.65	0.62	0.62
8. DLMN	0.40	0.55	0.57	0.53	0.59	0.65
9. DBLMS	0.40	0.54	0.57	0.67	0.61	0.61
10. DBTL	**0.50**	0.54	0.58	0.55	0.60	0.58
11. DBTMS	0.33	0.53	**0.65**	0.66	0.67	**0.71**
12. DBM	0.25	0.53	0.62	0.63	0.60	0.63
13. DBLN	0.45	0.52	0.63	0.55	0.64	0.60
14. DTLMS	0.35	0.52	**0.65**	0.61	0.68	0.66
15. DTLM	0.42	0.51	0.57	0.57	0.56	0.63
Causal	0.40	0.53	0.60	0.65	0.68	0.69
Causal − generic	**0.50**	0.55	0.64	0.69	**0.71**	0.69
T (E13)	0.23	0.43	0.44	0.50	0.56	0.53
BT (E13)	0.35	0.45	0.60	0.56	0.58	0.63
random	0.01	0.01	0.00	0.01	0.01	0.01

3 Hypotheses

H1 replication	Fails with the zero-shot patching of Yang App. G.1 (check 1: no gain on 7 tasks); the G.1 figures reproduce only with check 2 (wrong prompts, Sec. 2.4.1/App. H.1.3), which gains on all 14.	∼
H2 J-head ⊇ Logit-head	Fails: J-head misses part of the Logit-head set on 24/28 pairs. J-only heads are causal in both checks.	✗
H3 all-template heads	Falsified: better on ≤5/14 tasks.	✗
H4–H6 (E4, held-out)	Not decided (accuracies only, no intervals). Pattern: CC above E1 and random; FV above CC at every size except translations and prev-item at N_t.	–
H7–H9 (E4b)	Not run.	–

4 Design variants tried

Date	Tried first	Replaced by	Why / effect
10-27	Rules as main style, examples secondary	Rule and example treated alike	No main condition (user decision).
10-28	Prompts and means pooled over wordings	One wording for heads, one mean per wording (Yang's code)	Matches the source exactly.
10-28	Yang's batching without attention mask	Padding masked	Model attended to padding.
10-28	Check 1 only (zero-shot)	Check 1 + check 2 (released script)	Only check 2 reproduces Yang's figures.
10-30	First-token (simple) / strict (complex) score for heads and checks	Flexible score; E1, E2 rerun	Strict scoring created spurious wording differences.
11-02	E1b with $k = 5$ only	$k \in \{1, 3, 5\}$	After $k = 5$ failed; exploratory.
11-05	Yang's hand-written term lists	Lists from decoded words (E3), J-head only	Exploratory; light check 1 (α 1, 2, 4, 8).
11-05	Words and heads from the best template; 20-word lists	Both pooled over 5 templates; 5 + 5 words (CC-pooled)	0.18 → 0.45; shorter lists better.
11-05	Strict consensus at $\rho = 0.05$	$\rho = 0.15$	Run stopped and restarted (user decision).
11-05	Common words from task prompts	Words generic text elicits removed first (WikiText)	Helps facts, translations; hurts antonym, synonym, sequences.
11-06	Small sets by match rate	By attention to task words × decoding	User's L16H25 observation; larger effects at 5 and 20 heads.
11-06	E3: chosen and tested on the same items, best of 4 α	E4: held-out halves, 10 α (+ $\alpha = 1$), cross-fitted α	E3 designs were chosen on the test items. CC lead over E1 holds (0.45 → 0.44).
11-06	E4 bootstrap intervals and H4–H6 verdicts	Accuracy tables and curves only	User decision.
11-06	E4b not-named: no term-list word or task name	Only words naming the answer relation banned; input words (*country*, *landmark*) allowed	–

Inspecting replies: `notebooks/patching_outputs.py` (marimo) shows the simple-task patched replies for every manipulation: task, check, procedure (E1, $k = 1/3/5$, shared/unshared), readout, head and mean style, mean source and wording, α; accuracy, the share of the most common reply, reply kinds, and each item's replies.

5 What seems to work, what does not

- ✓ **Scoring matters first.** Flexible scoring removes most of the apparent wording sensitivity; always check it before reading wording effects.
- ✓ **Check 2 (wrong prompts, released-script setting)** is what reproduces Yang et al.'s figures, including those App. G.1 attributes to zero-shot patching; the zero-shot check works only on fact-like tasks.
- ✓ **J-head on simple tasks:** more heads, earlier layers (10–19), and the added heads carry the task causally. It does not contain the Logit-head set.

- ∼ **Which heads to keep** depends on the task type: all-template heads for fact tasks, union of templates for morphology-like tasks under J-head.
- ∼ **Retrieval heads** (heads that already write the answer token) are few among lexical task heads, but some effects rest on them: landmark–country's on its 10, present–past's union gain on L21H9. Elsewhere removing them changes little or helps (synonym, product–company, prev-item under J-head). Entity-answer tasks may be carried by retrieval more than by task heads.
- ∼ **Two roles on present–past** (one task): heads shared across tasks make the model *name* the task; L21H9 makes it give *a form of the input*. Either alone fails from a bare prompt; with an example in the prompt, L21H9 alone suffices.
- ✓ **J-head vs Logit-head at equal size:** J-head's larger effects mostly remain with 5, 10 or 20 heads per readout; Logit-head's few leads do not. Differences between template rules mostly come from set size.
- ✓ **Words the heads actually decode make better lists** (J-head, check 1): CC-pooled (5 task + 5 common words, words and heads over all 5 templates) is above E1 on all 14 tasks (mean 0.45 vs 0.19) and carries antonym, singular–plural, next-item and the translations, where E1 does nothing. Both word kinds and the pooling are needed.
- ∼ **Generic words help some tasks.** Removing the words heads read on any text (WikiText) helps facts, translations and morphology but hurts antonym, synonym and sequences.
- ∼ **Few heads suffice for named answer types:** heads that attend to the instruction's task words and decode the answer type carry capitals (1 head), landmark (5), product–company and singular–plural (20). Not translations, sequences, present–past.
- ✓ **E3 holds on held-out items** (E4): CC-pooled stays above E1 (0.44 vs 0.20) and random heads (0.02) with α chosen on other items. In check 2 the sets differ little.
- ✗ **Decoded words do not beat causal choice** (E4): at 20, 40 and N_t heads, heads ranked by Davidson's FV score carry most tasks better (present–past too); decoded-word sets are ahead only on translations and prev-item, at their own size.
- ✗ **Subtracting the instruction-free output does not sharpen the sets** (E6): heads chosen on what the instruction adds tie CC at 20 heads and fall far below it at full size in check 1; the heads whose words do not depend on the instruction are needed.
- ✓ **Attention separates working decoders from the rest** (E7): among heads that decode the task's most active words, keeping only those that look at the prompt rather than BOS cuts the set to about a quarter and keeps or raises check 1 (0.37 → 0.40); random subsets of the same size fall to 0.09. It matches causally chosen FV sets of the same size (FV-20/40); still below CC (landmark collapses) and FV-N_t; the WikiText cut halves the effect again.
- ✓ **Where a head looks, or what it decodes plus not looking at BOS, picks task heads** (E8): attention to the instruction's task words alone (0.55) and decoding the task's words with the BOS filter (0.51, no hand-written list) come close to FV-N_t (0.59) and win different tasks; attention to the query or the rest of the instruction hurts. Our lowercased lists had missed " Capital" tokens (E3–E7 lists still do).
- ✓ **One head set can serve all 14 tasks** (E10): ranking heads on E8's scores combined over tasks (average or average rank; DBT, BT, DB) gives 30 shared heads that carry check 1 about as well as E8's per-task sets (0.47 vs 0.49), better on facts, worse on antonym, synonym and translations; requiring every task to pass (weakest task) is worse.
- ✓ **Davidson et al.'s FV without its generic heads is the best set so far** (E11): one 25-head set for all tasks, added after layer 11 at $\alpha = 2$, above every per-task set in check 1 (0.63 vs FV-N_t 0.59); the same 30 heads patched head by head give 0.52, summed with the generic heads 0.39. Shared head-patched sets need about 30 heads; 20 are unstable across folds. Superseded by E12–E13: the generic heads were not chosen on the choosing half.
- ✗ **With generic heads chosen on the choosing half, the FV without them is no longer the best set** (E12): 0.63 → 0.49, mostly lost on translations, present–past and prev-item.
- ✓ **At equal size, attention filters match or beat Davidson et al.'s per-task ranking, and a shared causal ranking without generic heads is best overall** (E13): per task, DBT/BT reach their level at 40 heads (0.55), above Causal up to 60 heads and level at 80; shared, Causal − generic is best at every size (0.69 at 80). T is the strongest single filter and D adds nothing to BT; without T, larger DB sets make the model describe the query instead of answering. Generic heads hurt the causal sets and the summed FV but not DBT.
- ✗ **Causal heads do not read Bai et al.'s template or punctuation tokens more than the filter sets** (E20): what sets them apart is more attention to BOS and to the last token; the template-and-answer-slot profile belongs to the generic heads. Attention to the whole instruction, with no word list, picks heads that agree with the causal ranking as well as T does.
- ✓ **Attention to the instruction's punctuation and stopwords, with decoding and low BOS, is the best per-task rule at 40 heads** (E21): DBS 0.60 vs DBT 0.55; attention to the whole instruction alone (N, no word list) beats T (0.53 vs 0.45) and carries present–past. Heads that look at the last token or the Q/A markers carry nothing alone; shared, DB with those (DBM, DBL) comes close to Causal − generic, but nothing beats it.

- ✓ **Decoding and low BOS, plus attention to the instruction and the Q/A markers, is the best 40-head rule** (E22, all 79 combinations): per task DBMN 0.63, above every earlier per-task set at any size; shared DBTLS 0.60, above Causal − generic at 40 heads (0.55) but not at 80–100 (0.69–0.71). Decoding the task's words (D) is the letter that matters most; last-token and marker attention carry nothing alone and help only next to D and B. Picked as the best of 79 on the test items; E23 runs the 15 best per scope at every size (next).
- ✓ **Filter sets with stopword or marker attention are the best per-task sets at every size from 40 heads** (E23): DBTMS 0.68 at 100 heads, DBTLS 0.67 at 80, against Causal's 0.61 at 120; they win translations and most word relations, causal heads still carry sequences and morphology better. Shared, they match Causal − generic at its best (0.71) but beat it only at 40 heads. The 15 were picked on the test items at 40 heads, and the best size is read off the same items.
- **E3 caveats:** words, list sizes and designs chosen after seeing check 1 on the same items; best of 4 α; one run. E4 tests them on held-out items; new wordings (E4b) not yet run.
- **Read the replies, not only accuracy:** 19–33% of patched replies on several tasks contain the answer after a prefix the flexible score rejects, and part of a "loss" can be a format change (landmark–country under J-head). `notebooks/patching_outputs.py` shows them.

Not pursued: function vectors, function-specific heads, hidden-state differences (scope, 10-27); fp32 Head-JLens fit (≈136 GPU-h, declined).